

Witty Pi 4 Mini and the INIU Battery Power Bank

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I. INTRODUCTION

The Witty Pi 4 can be used with a battery power bank to save power when the Raspberry Pi is not in use. However, the INIU battery power bank has a smart feature that turns itself off when it detects it is not sourcing any power. This is not ideal because the button on the power bank will then need to be manually pressed to power the Raspberry Pi.

There are a few methods of keeping the battery bank awake, where some are easier than others. The main method tested on the INIU battery is a using a “dummy load”. A secondary method is also proposed.

II. DUMMY LOAD

A “dummy load” is when a load is connected to the power source with the purpose of keeping it awake. This can be implemented through the Witty Pi 4 built in “dummy load” or made externally.

A. Witty Pi’s Built-in Dummy Load

The Witty Pi 4 has time duration and cycle length settings for the “dummy load”. The time duration is from 0-254ms, and the cycle length is 1-25s. To check the feasibility of this method, the settings should be set at their most extreme values, so the battery has the best chance of staying awake. These settings are a time duration of 254ms and cycle length of 1s. After testing this feasibility, these settings could not keep the battery awake.

B. External Testing

The Witty Pi’s built in dummy load could not keep the bank awake due to limited settings. External testing was completed to get a better understanding of how the smart feature of the bank works.

Figure 1 shows the configuration for this external testing. A load resistor will be connected to the external contacts of the Witty Pi. A NMOS (will be connected to act as a switch, where it is controlled by a function generator. This is helpful because the load resistance, time duration, and cycle length can all be controlled with no limitations.

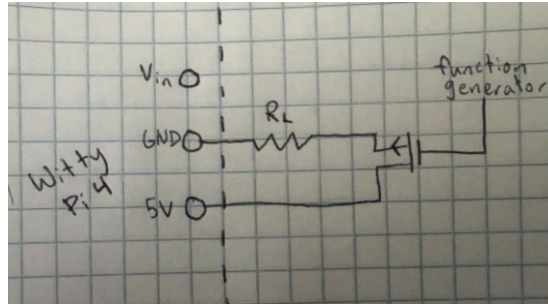


Figure 1: External Testing Configuration

There were different resistances, time durations, and cycle lengths that were tested. The most efficient setting was a 60Ω resistor, 2 seconds time duration, and 5 seconds cycle length. The average power consumption was around $0.2W$, which is not good for conserving energy. This is as good as just placing a permanent 120Ω resistor between GND and 5V.

It's still difficult to figure out how the INIU smart feature works, however it seems like a more complicated algorithm than usual, and a “dummy load” will most likely not work.

III. SECONDARY METHOD

The INIU power bank will stay on if the power button is repeatably “pressed”. This will also avoid any strange algorithms tied to the battery bank. If the contacts of the INIU power button is exposed, then an electronic switch can be connected so the button “presses” can be controlled by an electrical signal. This signal can come from the Witty Pi’s built in “dummy load”. Since the “dummy load” is connected to the batteries power button, but settings can most likely be very low and save more power.

However, this design was not tested and there is much public documentation for this. It also requires a soldering onto the battery pack and Witty Pi, which makes it a less of an ideal solution.

IV. CONCLUSION

The Witty Pi and INIU power bank have compatibility issues due to INIU’s smart feature. Through testing different “dummy load” settings, the INIU seems too resistant, and it will require a significant amount of power to stay awake. A secondary method was proposed which can potentially save a significant amount of energy, however the feasibility of it is unknown and it requires more effort to implement.